

Warrant

Maintenance records that are evidence, not paperwork. This document describes the system as it is actually deployed, one section at a time. Every value was read out of the live project with gcloud or the REST APIs, or by requesting the running service — not from the repository. Where the two disagree, the project wins and the node is marked drift.

● Live · 37 ● Provisioned · 6 ● Dormant · 3 ● Project · 1 ● On the bench · 3

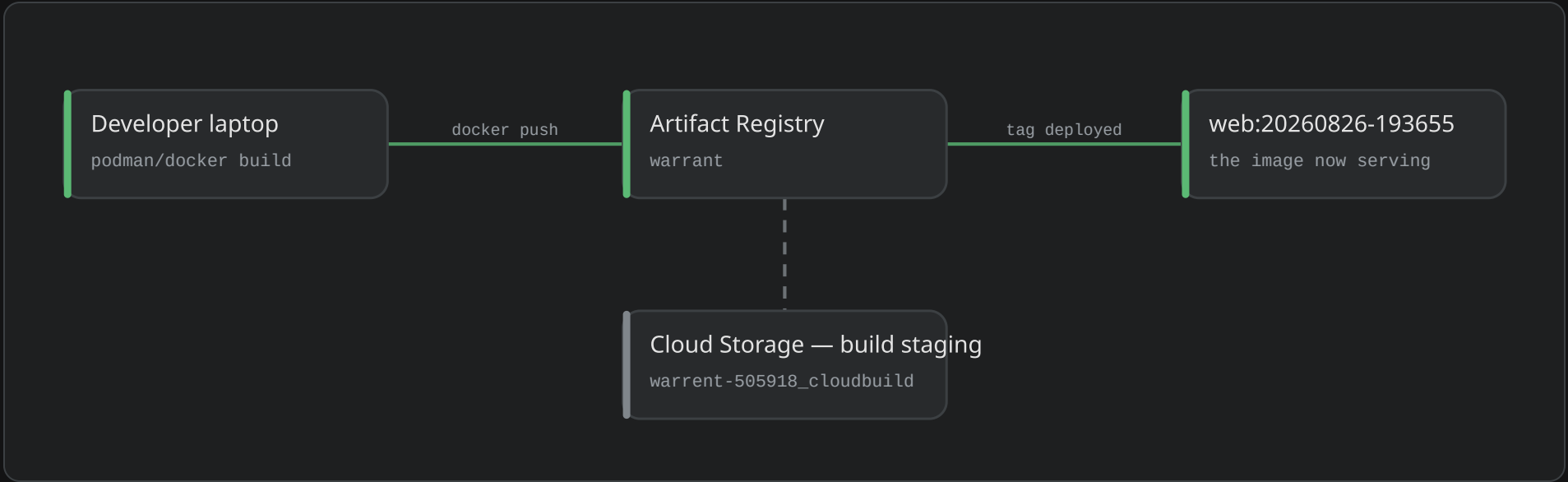
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Supply chain — how the image got there

The image is built on a laptop and pushed straight to Artifact Registry. Cloud Build is enabled and never runs.



● Developer laptop

podman/docker build · linux/amd64

ROLE	builds and pushes the image
WHY NOT CLOUD BUILD	new projects have no Cloud Build SA; <code>`builds submit`</code> fails PERMISSION_DENIED
SOURCE	infra/deploy-web.sh

The build host is outside Google Cloud entirely. **Cloud Build is enabled but never runs** — the image is built locally and pushed straight to Artifact Registry.

● Artifact Registry

warrant · Docker · us-central1

REPOSITORY	warrant
FORMAT	DOCKER
LOCATION	us-central1
IMAGES	web — 8 pushes, 3 tagged
MOST RECENT PUSH	2026-08-26 19:37 UTC

Every deploy this week came through here. Untagged layers accumulate because `deploy-web.sh` tags by timestamp and never prunes.

● web:20260826-193655

the image now serving

PUSHED	2026-08-26 19:37 UTC
BASE	node:22-alpine
BUILD	Next.js output: standalone
SERVING AS	revision warrant-00013-nlw

Built and deployed **today**. The tag is the build timestamp, so the image name is also the answer to “which commit is live”.

● Cloud Storage — build staging

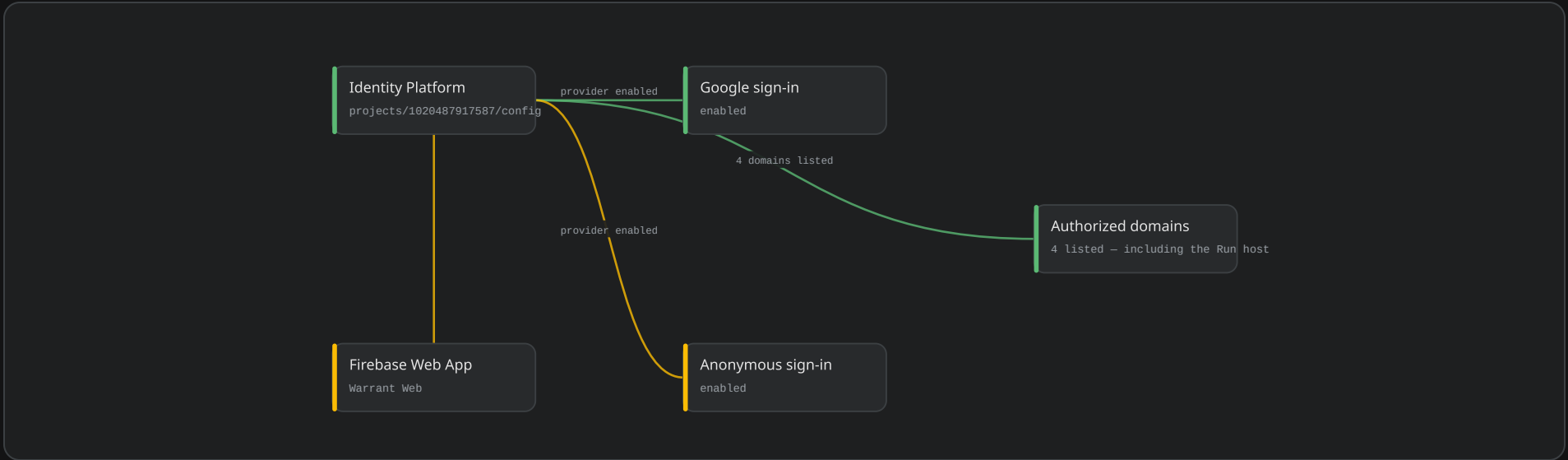
warrent-505918_cloudbuild · US

BUCKET	warrent-505918_cloudbuild
LOCATION	US multi-region
CREATED BY	Cloud Build API enablement
USED BY	nothing

Made by enabling an API that never ran. Left in place because deleting it is the sort of tidying that breaks a deploy at midnight.

Identity — Google sign-in, wired to the live origin

The browser never holds a long-lived credential: an ID token is exchanged server-side for an HttpOnly cookie.



● Identity Platform

projects/1020487917587/config

SERVICE	identitytoolkit.googleapis.com
PROVIDERS	google.com, anonymous
CALLBACK	warrent-505918.firebaseio.com/__/auth/action
SESSION	minted server-side into an HttpOnly cookie

The browser never holds a long-lived credential. The client SDK gets an ID token, /api/auth/session exchanges it, and everything after that is a cookie the page cannot read.

● Firebase Web App

Warrant Web · ACTIVE

STATE	ACTIVE
SUPPLIES	the apiKey and authDomain the client SDK needs
STORAGE BUCKET	warrent-505918-evidence

A registration, not a service. It exists so the browser SDK has something to point at.

● Google sign-in

enabled · OAuth client present

PROVIDER	google.com
ENABLED	true
CLIENT ID	1020487917587-9s2mkl1nncarhdu4c9uecn3s7rosfks
WHY MANUAL	no public API creates an OAuth client

The one piece of this project that cannot be created by a script. It was clicked once, by hand, in the console.

● Anonymous sign-in

enabled

ENABLED	true
USED BY	nothing yet
INTENDED FOR	a technician who joins a job by link

Enabled ahead of the flow that needs it. Nothing in the deployed app calls it.

● Authorized domains

4 listed - including the Run host

LISTED	localhost · warrent-505918.firebaseio.com · warrent-505918.web.app · warrant-zq2l2kwg3q-uc.a.run.app
WAS	3 - the Run host was missing
SYMPTOM WHEN MISSING	the sign-in popup is refused from the live origin

Fixed since the last reading. Sign-in worked on localhost and failed on the deployed site, which is the most expensive class of bug to find on the morning of a demo.

WHAT THIS SECTION REACHES OUTSIDE ITSELF

Identity Platform - /api/auth/* (Request path)

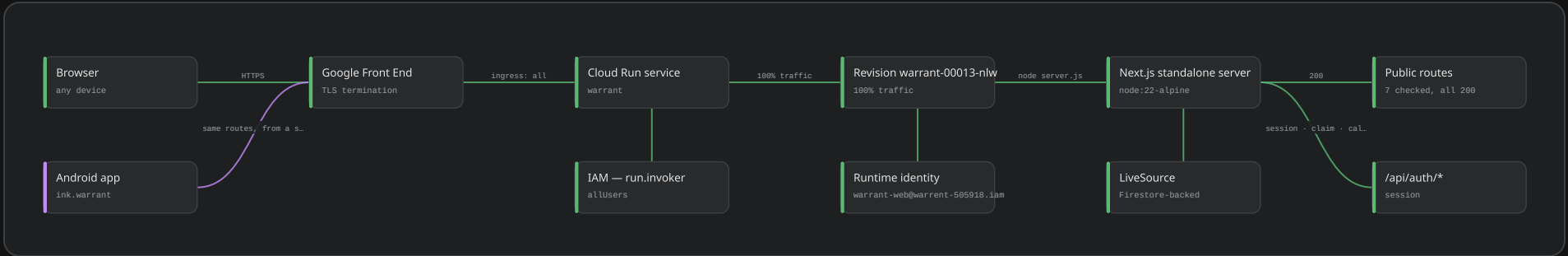
Identity Platform - warrant-web@ service account (Data)

Authorized domains - Google Front End (Request path)

Identity Platform - firebase-adminsdk-fbsvc@ (Data)

Request path — what a visitor actually touches

One container serves the pages, the API and the images. Every agent call is made from server code, never from the browser.



● **Browser**
any device · no account, no install

AUTH	Google sign-in, or none for public records
ENTRY	GET /
RESPONSE	200 in 0.41 s warm

The technician surface is a web page on purpose. **A shop cannot be asked to install anything** before it will try this.

● **Android app**
ink.warrant · Compose · debug build

TALKS TO	the same Cloud Run routes as the browser
ADDS	BLE instrument drivers the web cannot reach
DISTRIBUTION	sideloaded — no Play listing
SOURCE	android/app/src/main/java/ink/warrant

The second surface exists for one reason: **Web Bluetooth cannot be trusted to hold a GATT connection while a camera is open**. Everything else it does, the web does.

● **Google Front End**
TLS termination · global anycast

INGRESS	all
HOSTNAMES	warrant-zq2l2kwg3q-uc.a.run.app · warrant-1020487917587.us-central1.run.app
TO CONTAINER	plain HTTP :8080

Nothing was configured here. Cloud Run brings its own certificate and its own edge.

● **Cloud Run service**
warrant · us-central1 · managed

CONCURRENCY	80
SCALE	min 0 · max 4
TIMEOUT	300 s
STARTUP CPU BOOST	on
REVISIONS	13, all ready

Scale-to-zero is why the first request of the morning is slow and every one after it is not. **Max 4 instances is a cost ceiling**, not a capacity estimate.

● **IAM — run.invoker**
allUsers

ROLE	roles/run.invoker
MEMBER	allUsers
EFFECT	the service is open to the public internet

Public by design: a sealed record has to be readable by someone with the link and no account. **Authorisation happens above this line**, in the routes and in firestore.rules.

● **Revision warrant-00013-nlw**
100% traffic · 13th revision

CREATED	2026-08-26 19:37:54 UTC
TRAFFIC	100%, latestRevision
CPU / MEMORY	1 vCPU · 512 MiB
PORT	8080 (http1)
READY	true

Thirteen revisions in seven days. Traffic always follows latest — there is no canary here, and a bad deploy is fixed by deploying again.

● **Runtime identity**
warrant-web@warrent-505918.iam

ATTACHED SA	warrant-web@warrent-505918.iam.gserviceaccount.com
WAS	the default compute SA with roles/editor, until 2026-08-21
CREDENTIALS	metadata server, short-lived
REACHES VERTEX BY	impersonating warrant-adjudicator@

Fixed since the last reading of this diagram. The service ran as the default compute account — project editor — for its first two revisions. It now runs as a purpose-made account that cannot administer the project it lives in.

● **Next.js standalone server**
node:22-alpine · server.js on :8080

MODE	output: standalone
PROCESS	node server.js, PID 1
STATIC	/_next/static served in-process
ROUTES	22 API routes · 17 pages

One container serves the pages, the API and the images. There is no separate backend, and **every agent call in this system is made from server code**, never from the browser.

● LiveSource

Firestore-backed · fabricated = false

BINDING

REASON

WAS

FALLBACK

getDataSource() → LiveSource

GCP_PROJECT and the Firebase env are baked into the image

FixtureSource, until the env was set on 2026-08-21

FixtureSource still runs the whole product offline

Fixed since the last reading.

The deployed image used to fall back to fixtures because no configuration reached it, and the site said so out loud. It now reads and writes the real database.

● Public routes

7 checked, all 200

CHECKED

200

RENDERER

SLOWEST

2026-08-26, live service

/ · /library · /model-tests · /about · /author · /fleet · /records

React Server Components + client islands

/fleet, 1.87 s — it queries the engine

Every page a judge is asked to open was requested from the public internet on the day this diagram was made.

● /api/auth/*

session · claim · calendar

/API/AUTH/SESSION

/API/AUTH/CLAIM

WAS

ALSO

200

405 — POST only, as written

all 404 — absent from the image on 2026-08-19

/api/auth/calendar for the OAuth grant

Fixed since the last reading.

These routes existed only in the working tree at the time of the last deploy, so sign-in could not complete against the live service.

WHAT THIS SECTION REACHES OUTSIDE ITSELF

Revision warrant-00013-nlw — web:20260826-193655 (Supply chain)

LiveSource — Firestore (default) (Data)

Google Front End — Authorized domains (Identity)

Cloud Run service — 51 APIs enabled (Platform services)

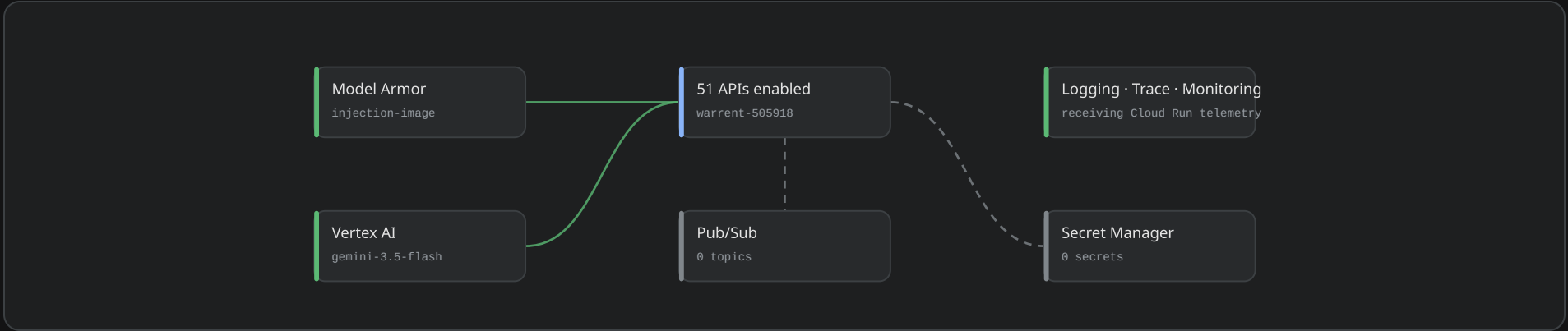
Revision warrant-00013-nlw — Logging · Trace · Monitoring (Platform services)

/api/auth/* — Identity Platform (Identity)

Public routes — Vertex AI Agent Engine (The fleet)

Platform services

What the project actually calls, and what was enabled and left alone. Both are drawn, because an unused service is still a fact about the build.



● Model Armor

injection-image · locations/us

TEMPLATE projects/warrent-505918/locations/us/templates/injection-image

CREATED 2026-08-18

FILTERS prompt injection · jailbreak · RAI DANGEROUS at LOW_AND_ABOVE

WIRED BY MODEL_ARMOR_LOCATION + MODEL_ARMOR_TEMPLATE on the revision

Now in the request path. Every capture is screened before a judgement model is asked about it, because a **photograph is untrusted input** and a sticker with words on it is a prompt.

● Vertex AI

gemini-3.5-flash · global endpoint

JUDGEMENT MODEL gemini-3.5-flash

SCREENING MODEL gemini-3.5-flash-lite

ENGINE-SIDE gemini-3.5-flash-lite, observed serving 2026-08-26

LOCATION global — the model is not where the rest of this runs

TEMPERATURE 0.0

Two tiers, chosen by **what it costs to be wrong**: Flash-Lite refuses an unusable photo, Flash judges the evidence, and nothing cheap can pass a step. **The screen is not Gemma** — Gemma is not a publisher model on Vertex and every spelling of it 404s, so running it would mean paying for a GPU endpoint that has to still be up when a judge watches the film.

● 51 APIs enabled

warrent-505918

ENABLED 51

WAS 46 on 2026-08-19

ENABLED BY infra/bootstrap.sh, once

Most of these are transitive — enabling Firebase pulls in a dozen. The ones this system actually calls are drawn.

● Pub/Sub

0 topics

TOPICS 0

SUBSCRIPTIONS 0

ENABLED BY bootstrap.sh

Enabled and never used. The sweep is a cron, not a queue — see the scheduled band.

● Logging · Trace · Monitoring

receiving Cloud Run telemetry

LOGGING request and stdout logs

TRACE spans named agent.inspector, agent.skeptic, screen — the screen runs Flash-Lite

MONITORING Run built-in metrics

GAP the app logs no stack on a 500

The agent calls are traced by hand, so a slow adjudication can be read span by span. **Application errors are not logged**, which is how the broken sweep stayed quiet.

● Secret Manager

0 secrets

SECRETS 0

WHERE THE SECRETS ARE INSTEAD plain env vars on the revision

AT RISK the sweep secret, the OAuth client secret, the instrument keys

The honest weak point of this deployment. Cloud Run env vars are visible to anyone with describe on the service. For a hackathon build that is a considered trade; for a real fleet it is the first thing to move.

WHAT THIS SECTION REACHES OUTSIDE ITSELF

Logging · Trace · Monitoring — Revision warrent-00013-nlw (Request path)

Vertex AI — The cheap screen (The fleet)

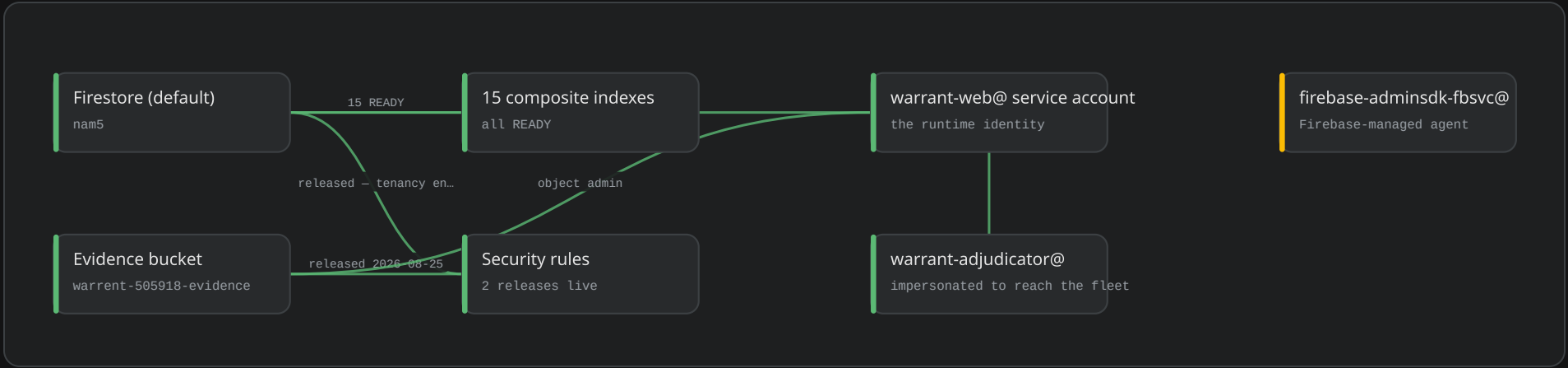
51 APIs enabled — Cloud Run service (Request path)

Vertex AI — Vertex AI Agent Engine (The fleet)

Model Armor — Vertex AI Agent Engine (The fleet)

Data — carrying real records

Everything lives under tenants/{t}/..., so a query that forgets its tenant returns nothing rather than everything.



● **Firestore (default)**

nam5 · Native · pessimistic

TENANTS	22
JOBS	35
PROCEDURES	56 (+1 published publicly)
SEALED RECORDS	2
MEMBERS	21

No longer empty. Everything is under tenants/{t}/... so a query that forgets its tenant returns nothing rather than everything. The two sealed records are the ones the demo was cut from.

● **15 composite indexes**

all READY

COMPOSITE INDEXES	15
WAS	7 on 2026-08-19
DEPLOYED BY	infra/deploy-rules.sh

Every one of these exists because a query failed once. Firestore names the index it wants in the error, which makes this the least interesting file in the repository and the most annoying to omit.

● **Security rules**

2 releases live

CLOUD.FIRESTORE	released 2026-08-25 14:11 UTC
WARRENT-505918-EVIDENCE	released 2026-08-25 14:11 UTC
WAS	2 rulesets uploaded, neither released
PROVED BY	the emulator, in scripts/smoke.sh

Fixed since the last reading. Rules that are uploaded but never released do nothing at all, and the console shows them as if they were live. Tenancy is now enforced by the database, and smoke.sh proves it by running the rules rather than asserting

● **Evidence bucket**

warrent-505918-evidence · 52 objects

OBJECTS	52
LOCATION	US-CENTRAL1
HOLDS	capture media, avatars, uploaded paper forms
RULES	released 2026-08-25

The photographs themselves. There was no storage rule at all until 2026-08-20, while the bucket was already configured — the default posture must not be discovered on the day the first photograph is uploaded.

● **warrant-web@ service account**

the runtime identity

ATTACHED TO	the Cloud Run revision
HOLDS	datastore.user · firebaseauth.admin · storage object admin on the evidence bucket
DOES NOT HOLD CAN	project editor impersonate warrant-adjudicator@

Least privilege, and now actually in use. It cannot administer the project, cannot create service accounts, and cannot read another project's data.

● **warrant-adjudicator@**

impersonated to reach the fleet

ROLE	aiplatform.user, and nothing else
REACHED BY	warrant-web@ minting a short-lived token
WHY SPLIT	the web identity should not hold a standing key to the model

The split exists so that the credential that can call a judgement model is not the credential serving web pages. It is issued for minutes at a time and never written down.

● **firebase-adminsdk-fbsvc@**

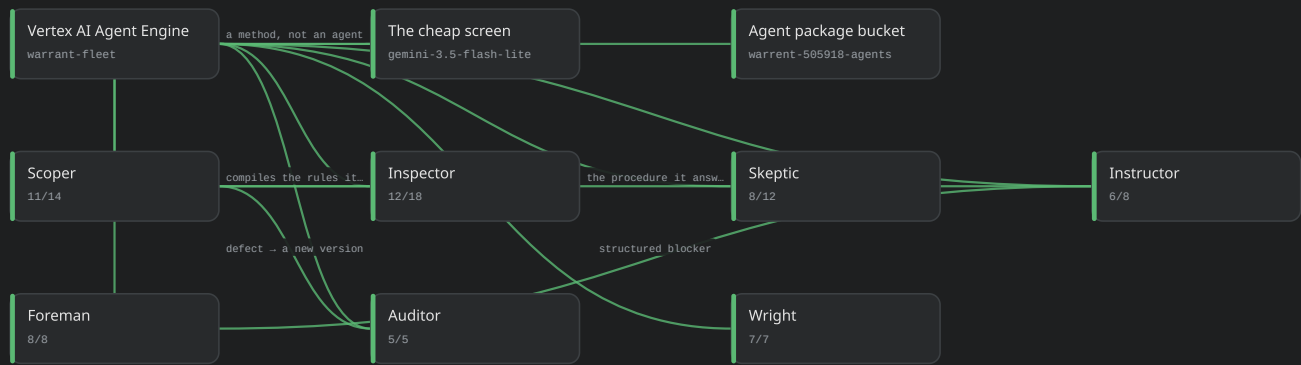
Firebase-managed agent

CREATED BY	Firebase, automatically
USED BY	nothing in this deployment

Comes with the Firebase registration. Left alone.

The fleet — deployed on Vertex AI Agent Engine

Seven agents behind one engine, which answers for its own roster over the API — so the claim is checkable from outside.



Vertex AI Agent Engine

warrant-fleet · reasoningEngines/5032906174249304064

DISPLAY NAME	warrant-fleet
CREATED	2026-08-21 22:16 UTC
LAST UPDATED	2026-08-25 16:03 UTC
METHODS	query · roster · screen
ROSTER ANSWERS	seven agents, each with its contract

The fleet is deployed, not on a laptop. Asked for its roster over the API it names all seven agents and the contract each one answers under — so the claim on this diagram is checkable from outside by anyone with the endpoint.

The cheap screen

gemini-3.5-flash-lite · not an eighth agent

ANSWERS	UNUSABLE or NEEDS JUDGEMENT — there is no PASS
COSTS WHEN WRONG	one retake
NOT IN	REGISTRY — roster() says seven
WHY NOT GEMMA	not a publisher model on Vertex — every spelling returns 404
TRACED AS	screen

The cheap model sits only on the side where being wrong is recoverable. It has no PASS in its enum, so the strongest answer it can give still ends with the judge or the technician being asked. That is a shape,

Agent package bucket

warrent-505918-agents · 13 objects

OBJECTS	13
HOLDS	the packaged fleet the engine runs
WRITTEN BY	infra/deploy-agents.py

How the code in agents/ gets to Vertex. The engine pulls from here at deploy, not at request time.

Scoper

11/14 · interviews, then compiles

DECIDES	what to ask next, and when a procedure would run unambiguously
CONTRACT	scoper-turn.schema.json
SCENARIOS	14, including full interviews
REACHED BY	/api/scoper/turn and /api/procedures/compile
REFUSES	to state a tolerance nobody gave it

There is no form builder in this product and this agent is the reason. Its hardest scenario runs a whole interview against a shop played by a second model, then checks every bound in the compiled procedure against the numbers the shop actually...

Inspector

12/18 · one field at a time

DECIDES	PASS · ADD FIELD · ESCALATE on one field's evidence
CONTRACT	inspector-verdict.schema.json
SCENARIOS	18 (4 need media not checked in)
REACHED BY	/api/adjudicate
NEVER JUDGES	a whole step — it would trade a weak photo against a strong one

Reads the photograph against the written acceptance rule, and where it falls short composes the specific next request rather than asking for a generic retry. Numeric bounds are not its job: within(min,max) is deterministic code, and the...

Skeptic

8/12 · adversarial

DECIDES	does this evidence belong to this job, this machine, this moment
CONTRACT	skeptic-verdict.schema.json
SCENARIOS	12 (3 need media not checked in)
REACHED BY	/api/adjudicate, beside the Inspector
NEVER SEES	the Inspector's verdict — its prompt does not mention an Inspector exists

Given the perceptual embedding distance as an input: code computes the distance, the model decides what it means against the machine's recorded damage and history. Twelve identical bikes

Instructor

6/8 · unbounded speech in

DECIDES	the blocker class, the next action, and the safety flag
CONTRACT	instructor-recommendation.schema.json
SCENARIOS	8
REACHED BY	the disposition path, when a step stalls
SEES	the technician's words verbatim, untidied

"I haven't got the right socket" and "the socket won't bite, it's rounded" share a vocabulary and mean tool_missing versus seized_or_damaged — which decides whether the next visit needs an extractor or

Foreman

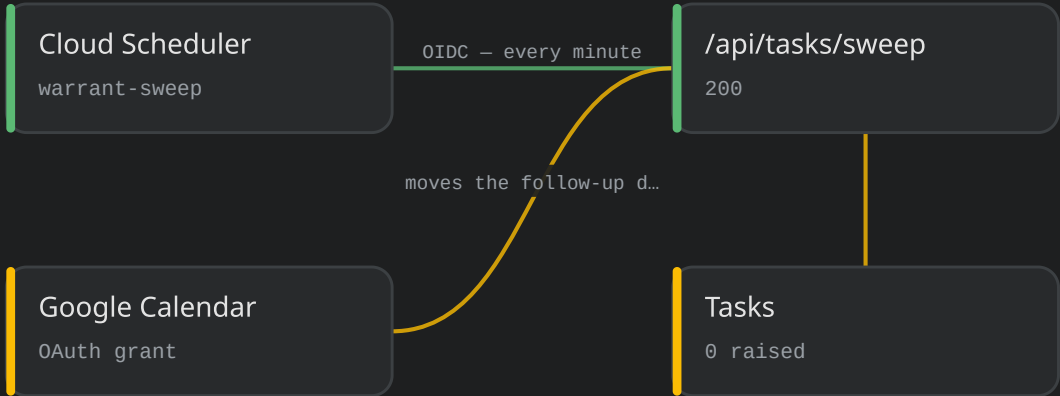
8/8 · owns the job for its life

DECIDES	status, action, and whether to recommend holding the machine
CONTRACT	foreman-disposition.schema.json
SCENARIOS	8
REACHED BY	the disposition path and the sweep
CANNOT	hold anything itself — the Gate does, deterministically

Its inputs conflict on purpose: the machine is unsafe, the customer is booked in two days, and the torque wrench arrives Thursday. It writes down enough to be woken days later with no other context. A waiver needs a named person holding standing, and...

Scheduled work — one cron for the whole system

One cron re-reads the world every five minutes. It is currently failing, and nothing downstream of it has ever run.



● Cloud Scheduler

warrant-sweep · * * * * *

SCHEDULE	* * * * * — every minute
WAS	* /5, until deploy-web.sh reset it
STATE	ENABLED
TARGET	POST /api/tasks/sweep
AUTH	OIDC token

One cron for the whole system. Cloud Tasks would be more precise; a cron that re-reads the world cannot lose work when a deploy lands mid-flight.

● Google Calendar

OAuth grant · /api/auth/calendar

SCOPE	calendar events
WRITTEN BY	the sweep
EVENTS CREATED	none

A held machine that needs a part on Thursday should put Thursday in somebody's calendar. Wired, granted, never exercised.

● /api/tasks/sweep

200 · 3–11 s

STATUS	200 on 2026-08-27, revision 00014
WAS	500 on every run for a day, across revisions 00012 and 00013
RUNTIME	10.6 s cold, 3.0 s warm
UNAUTHENTICATED	401, correctly
HOLDS	a lease, so a long run is not overtaken by its own retry

This was the one red node on this diagram, and it is fixed. For a day the cron authenticated, threw early and returned 500 in under a second — before any model call, on two revisions, so no single deploy caused it. Revision 00014 answers 200 and completes in seconds. Two consecutive runs observed; the collections it writes are still empty, which is a separa...

● Tasks

0 raised

TASKS	0
RAISED BY	the Foreman and the Auditor, through the sweep
NOTE	the sweep now completes, so an empty table is a real answer

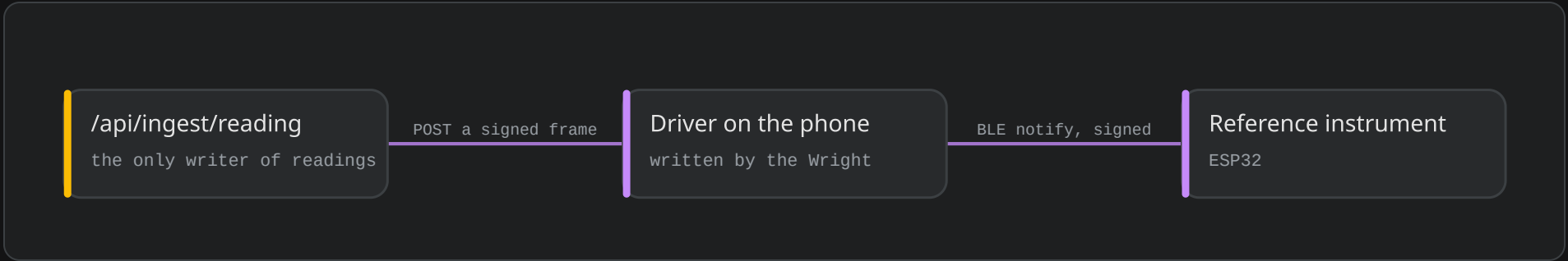
Still empty — but the sweep above now runs, so this is no longer explained by a broken cron. Two sealed records is a thin window for an Auditor whose subject is a procedure across weeks of jobs; there is genuinely little for it to find yet.

WHAT THIS SECTION REACHES OUTSIDE ITSELF

/api/tasks/sweep → Auditor (The fleet)

Instruments — a reading nobody typed

A signed frame from an instrument is the difference between a measurement and a number somebody typed.



● Reference instrument

ESP32 · BLE · signed frames

TRANSPORT	BLE GATT notify
SIGNS	every frame, with a counter persisted in NVS
WHY A COUNTER	BLE is a broadcast — without it a frame can be replayed
SOURCE	firmware/warrant_reference_instrument.ino

Real hardware on a bench, not a simulation. It exists to prove the shape: **a torque wrench that signs its readings** means the number on the record was measured rather than typed.

● Driver on the phone

written by the Wright

READS	the GATT tree, then the characteristic the Wright chose
RUNS ON	the Android surface only
FALLS BACK TO	a typed value, recorded as unattested

The phone is the bridge. An instrument that cannot be read is not a blocker — the reading is simply recorded as one a person typed, and the record says so.

● /api/ingest/reading

the only writer of readings

READINGS WRITTEN	0 in production
RULES	refuse `readings` from every client
UNSIGNED READING	accepted, but recorded with no tool_id
KEYS	WARRANT_INSTRUMENT_KEYS, compared in constant time

The **only writer of readings**, and that is the point of the endpoint. No client may write a measurement directly, so “measured” means a signed frame reached this route — and an unsigned one still gets recorded, just without the claim.

One capture, end to end

The sections above are the parts. This is the path a single photograph takes through all of them, in order, on the deployed system.

- 1

A technician opens the job on the web or the Android app. There is nothing to install, and a sealed record is readable by anyone with the link — which is why the service is public to allUsers.
- 2

Google Front End terminates TLS and hands plain HTTP to the Cloud Run service warrant, one container serving the pages, the API and the images.
- 3

The revision runs as warrant-web@ — not the default compute account it used to run as — so the identity serving web pages cannot administer the project.
- 4

The capture is written to the evidence bucket and the job to Firestore, both under tenants/{t}/..., with released rules refusing anything that reaches across a tenant.
- 5

Model Armor screens the photograph first. A photograph is untrusted input, and a sticker with words on it is a prompt.
- 6

The cheap screen (Flash-Lite) asks one question: is this frame so unusable that spending the judge on it is waste? It has no PASS in its enum, so it can cost a retake and nothing more.
- 7

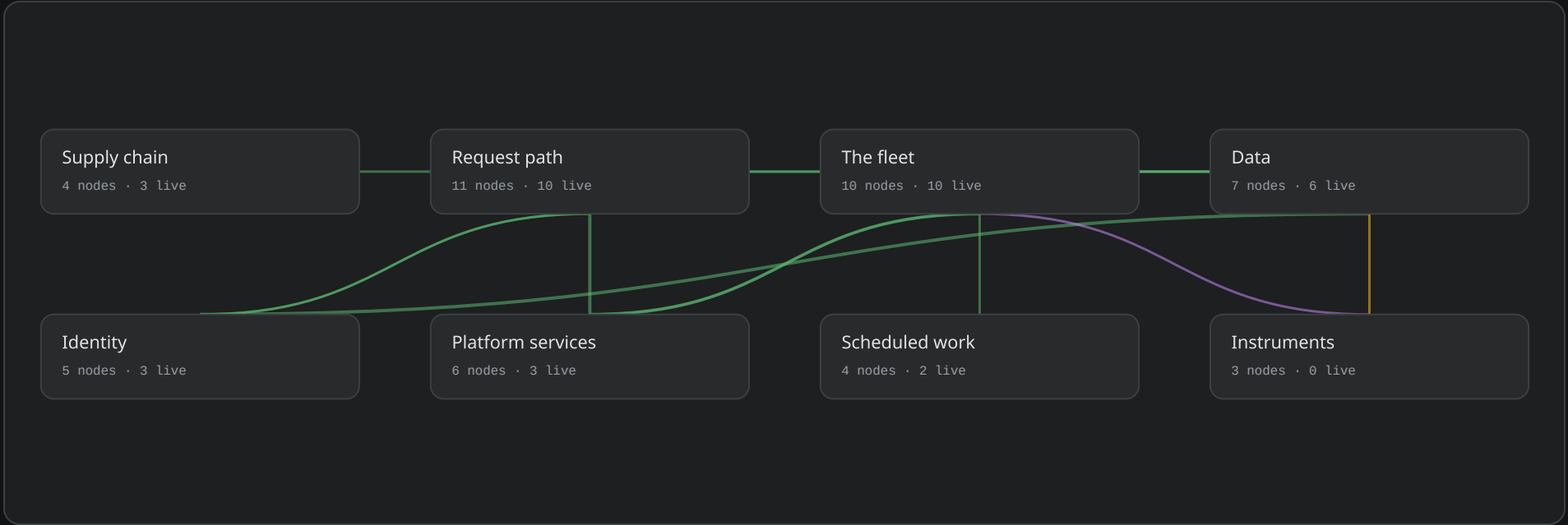
warrant-web@ mints a short-lived token for warrant-adjudicator@ and calls reasoningEngines:query. The credential that can reach a judgement model is never the one serving pages.
- 8

The Inspector judges the evidence against the written rule; the Skeptic asks whether it belongs to this job at all. **Neither sees the other's verdict** — two agents shown each other's conclusions agree, and the second opinion stops being one.
- 9

The verdict lands on the record with the model and the version that produced it. The Gate — deterministic code, not an agent — decides whether the step passes and whether the machine is held.
- 10

Every five minutes Cloud Scheduler calls the sweep, which is where the Auditor reads weeks of jobs at once and hands a procedure defect back to the Scoper.

THE EIGHT SECTIONS, AND THE TRAFFIC BETWEEN THEM



Step 10 was broken for a day, and is not any more. The sweep returned 500 on every run across two revisions, failing in under a second — before any model call — so no audit ran against real data. Revision 00014 answers 200 and completes in three to eleven seconds. tasks, findings and audits are still empty, which is now a thinness-of-data question rather than a broken one: two sealed records is a narrow window for an Auditor whose subject is a procedure across weeks of jobs.